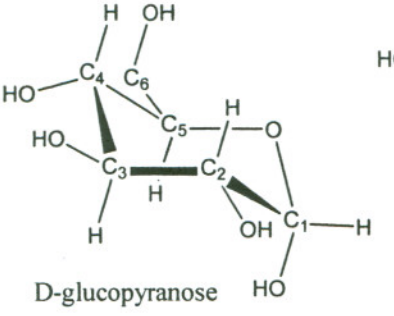
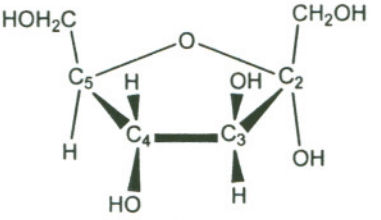
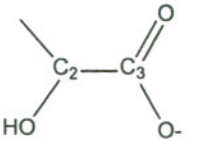
Biochemistry Cume
2/11/11



D-glucopyranose



D-fructofuranose



lactate

1. (50 pts) Fructose labeled with ^{14}C at C-2 is incubated with glycolytic enzymes and necessary cofactors. Hexokinase can also convert D-fructose to fructose 6-phosphate. Which carbon atom(s) of fructose 1,6-bisphosphate, dihydroxyacetone phosphate, 1,3-bisphosphoglycerate and pyruvate would be labeled? Give metabolic pathways for these reactions, and show your work.

2. (50 pts) Lactate labeled with ^{14}C at C-2 is added to a cell extract containing the enzymes and cofactors of the glycolytic pathway, the citric acid cycle, and the pyruvate and lactate dehydrogenase complexes. Draw the correct structure and show labeled carbons in the following intermediate molecules: pyruvate, succinate, acetyl-CoA, dihydroxyacetone phosphate. Include the names or abbreviations of any other reactants and products (eg, ATP, ADP, Pi, NAD+, NADH, etc).